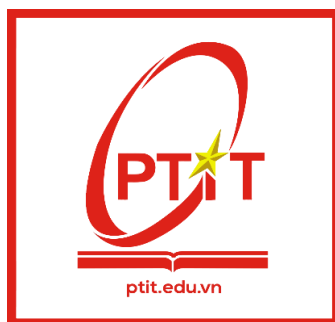


**POSTS AND TELECOMMUNICATIONS INSTITUTE OF
TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY**



**TERM PROJECT REPORT:
Business Intelligence
Topic:
SocialLens BI**

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PREFACE

This report presents the final result of SocialLens BI, a Business Intelligence project developed for the Business Intelligence course. The project focuses on analyzing social media performance in the Vietnamese F&B market, with Highlands Coffee Vietnam selected as the primary brand and several competitors used for benchmarking.

The original idea of the project was broader, covering multiple social media platforms and a larger set of analytical features. During implementation, however, the scope was narrowed to official YouTube channels. This decision was made after evaluating API access constraints, source reliability, and the risk of mixing official brand content with noisy query-based data. As a result, the final system prioritizes data quality, traceability, and defensible metrics over raw data volume.

The report describes the complete BI workflow: data extraction from YouTube Data API v3, ETL processing, PostgreSQL data warehouse design, analytical views, Django API, Next.js dashboard, validation checks, and business insights. The final dataset contains 817 videos, 1,526 sentiment-classified comments, and covers the period from 2017-09-26 to 2026-05-26.

The purpose of this report is not only to show dashboard results, but also to explain how the data was collected, cleaned, modeled, validated, and interpreted. This is important because BI outputs are only useful when the underlying definitions, assumptions, and limitations are clear.

Project repository: https://github.com/phmiinh/Nexa_BI

Chapter I: INTRODUCTION AND OBJECTIVES

1. Research Background

The F&B sector in Vietnam is a highly competitive market characterized by short campaign cycles and a strong reliance on brand perception. Brands such as Highlands Coffee, Phúc Long, Cộng Cà Phê, and Trung Nguyên Legend compete not only on the basis of store count, pricing, or product offerings, but also on their capacity to sustain a presence in consumer awareness through brand storytelling, seasonal campaigns, promotions, aesthetic imagery, community sentiment, and digitally distributed campaign videos.

In practice, social media analytics are frequently fragmented across channel-specific reports, campaign-level documents, individual CSV files, or platform-native dashboards. While adequate for reviewing isolated metrics, this approach falls short of answering comparative BI questions: which brand leads in reach, which generates more meaningful discussion, which content formats convert views into engagement, whether sentiment poses any risk, and whether the data meets the quality standards required for management-level reporting. SocialLens BI was developed to translate these questions into a structured analytical workflow.

The first objective of the project was to construct an operational data pipeline sourced from real data. This criterion is significant because a visually polished dashboard built on mock data fails to reflect the core challenges of BI: real-world data typically contains missing fields, inconsistent formatting, API rate limits, duplicate keys, timezone discrepancies, and noise from non-official sources. The final version of SocialLens BI uses the YouTube Data API v3 and PostgreSQL as the source of truth and does not rely on static JSON or sample data in the runtime dashboard.

The second objective was to design a data warehouse suited for analytical purposes. Rather than having the dashboard query raw files directly or call social media APIs on each page load, the team constructed the `social_dw` schema following the star schema model. Fact tables store video performance and comment sentiment data; dimension tables normalize time, platform, content type, and page/channel attributes. An analytical view layer transforms granular data into datasets readily consumable by the dashboard, API, and export modules.

The third objective was to generate business-relevant insights. The project extends beyond simple post counting. The report aims to surface actionable findings: Highlands commands a large reach but has a disproportionately low engagement rate; Trung Nguyên Legend serves as a strong benchmark for content cadence; overall sentiment is safe but predominantly neutral; the posting time heatmap requires interpretation through both volume and engagement rate lenses; and sentiment analysis for brands with small comment samples should not be over-interpreted.

2. Business Questions

The project is structured around three primary business questions. These questions are sufficiently specific to be translated into KPIs, database views, and dashboard pages, while remaining broad enough to inform content governance recommendations.

- What content performs effectively? Which video formats, messaging approaches, posting times, and content categories generate the highest engagement for Highlands Coffee and its competitors?
- How is the audience responding? Do comment sentiment distributions trend positive, neutral, or negative, and are there any risk signals of sufficient magnitude to warrant monitoring?
- Where does Highlands stand relative to its competitors? How does Highlands' share of voice, engagement rate, post volume, and comment coverage compare to Trung Nguyên Legend, Phúc Long, Cộng Cà Phê, and the remaining brands?

From these three questions, the dashboard addresses more granular operational inquiries: what distinct narratives do reach/views, engagement, engagement rate, and share of voice each convey; which individual videos generate notable interaction; which posting time slots merit further experimentation; and whether the data is sufficiently clean and traceable for use in the final academic submission.

3. Selected KPIs and Rationale

The selected KPIs are organized around four dimensions: awareness performance, engagement performance, sentiment health, and competitive positioning. In this report, reach is a proxy derived from YouTube views/impressions and does not represent unique reach. share_of_voice is reach-based within the approved official

channel set and is not mention-based across the broader market. Explicitly defining these terms is essential, as the same KPI name can carry different meanings depending on the data source.

KPI	Formula	Rationale	Scope Notes
engagement_count	likes + comments + shares + saves	Measures total available interactions at the video level	In the YouTube final dataset, shares and saves are unavailable and equal 0
engagement_rate	engagement_count / reach * 100	Normalizes engagement relative to reach/view volume	Used for cross-channel comparison between large and small accounts
virality_score	shares / reach * 100	Measures virality potential when share data is available	Equals 0 in this snapshot because the YouTube API does not expose share count
sentiment_ratio	positive_comments / total_comments * 100	Tracks emotional health through comment sentiment	Should only be interpreted meaningfully when the comment sample is sufficiently large
share_of_voice	page reach / total scoped reach * 100	Measures competitive positioning	Represents reach-based SOV, not mention-based SOV across the full market

		within the official channel set	
--	--	---------------------------------------	--

4. Deliberate Scope and Limitations

The final project scope is limited to the official YouTube channels of 8 F&B brands, covering data from 2017-09-26 to 2026-05-26. This scope excludes Facebook, TikTok, Instagram, and earned media, as these sources either pose API access risks or lack a sufficiently robust official source verification mechanism within the capstone timeframe. The Coffee House was also excluded from the clean warehouse because a reliable official YouTube channel ID could not be confirmed.

These are deliberate limitations, not concealed deficiencies. The project prioritizes a smaller but cleaner pipeline — one with natural keys, quality gates, and interpretable outputs. In the context of BI, a benchmark with fewer sources but precise definitions is generally more defensible than a dashboard with a large number of sources that conflates official content, user-generated content, and noisy search-based data.

Chapter II: METHODOLOGY

1. Dimensional Modeling – Kimball Theory

Business Intelligence is not simply about visualization. A well-designed BI system must be capable of addressing three analytical tiers. The first tier concerns what occurred: post count, views, engagement totals, and sentiment distributions. The second tier addresses why it is noteworthy: high reach combined with low engagement, large content volume but a small share of voice, or neutral sentiment significantly exceeding positive sentiment. The third tier identifies recommended actions: separating awareness and engagement KPIs, experimenting with posting time slots, prioritizing calls-to-action to increase positive comments, and refraining from expanding data sources before source confidence is established.

The team adopted a warehouse-first approach. This means that data after extraction does not flow directly to the dashboard; it passes through a normalization and validation layer first. This introduces additional upfront design costs but ensures that the final report's figures are defensible. When the dashboard reports a total of 817 posts, that number is not a transient API result — it resides in a fact table, has passed quality gates, and has been reconciled against analytical views.

In theoretical terms, the Kimball methodology emphasizes organizing data around business processes and designing around a clearly defined grain. In this project, the primary business process is the activity of publishing videos and receiving comment interactions on official YouTube channels. Two fact tables were accordingly defined:

- fact_post: grain is one video/post per platform.
- fact_sentiment: grain is one sentiment-classified comment.

A star schema was selected instead of a snowflake schema because the analytical workload of this project is dashboard-oriented and read-heavy. Most queries aggregate metrics by page, time, platform, or content type; keeping these dimensions directly connected to the fact tables makes the model easier to query, easier to explain, and more suitable for BI visualization. A snowflake design could further normalize attributes such as calendar hierarchy, platform metadata, or page classification, but that would introduce additional joins without adding meaningful

analytical value for the current dashboard scope. For this project, clarity and query simplicity are more important than maximum normalization.

Dimension tables are designed to support slicing and dicing: `dim_time` for analysis by day, hour, week, and month; `dim_platform` for platform-level segmentation; `dim_content_type` for content format classification; and `dim_page` for brand and competitor identification. This design is more appropriate for BI purposes than a single denormalized flat table, as dashboards typically require cross-dimensional comparisons by page, time period, content type, and competitor flag.

A fundamental principle of dimensional modeling is maintaining grain consistency. If `fact_post` represents a single video, then metrics such as reach, likes, comments, and `engagement_count` must be understood at the video level. If `fact_sentiment` represents a single comment, then sentiment ratio should not be stored directly in the comment fact table but should instead be computed through view-level aggregation. The `social_dw` schema adheres to this principle throughout.

2. ETL Pipeline — Concepts and Implementation

ETL in this project is understood in its full operational sense: Extract, Transform, and Load. Extraction invokes the YouTube Data API v3 and persists raw JSONL files as an auditable source record. Transformation normalizes payloads, converts timestamps, maps content types, computes sentiment classifications, and produces processed CSV tables. Loading ingests data into PostgreSQL via upsert operations, ensuring that repeated batch runs do not produce duplicate rows.

The distinction between a toy ETL exercise and production-grade ETL lies in the less elegant aspects: API quota exhaustion, 403 errors on comment retrieval, difficulty verifying official channel identities, noisy results from query-based searches, timezone normalization requirements, conventions for handling missing metrics, and the risk that sample fallback data may inadvertently corrupt reporting outputs. For this reason, the team chose to fail explicitly when production configuration or API access is unavailable, rather than silently substituting sample data. This represents a deliberate prioritization of BI correctness over operational convenience.

3. Sentiment Analysis in Vietnamese NLP

Sentiment analysis is a considerably more complex NLP task than counting likes or views. For the Vietnamese language, the challenge is amplified by the prevalence of slang, missing diacritical marks, code-switching, emoji usage, sarcasm, and product-specific contextual nuances. An ideal implementation might employ PhoBERT or a transformer-based model fine-tuned on a labeled Vietnamese sentiment dataset. However, within the constraints of the capstone environment, the team opted for a lightweight Vietnamese rule-based fallback approach.

This decision involves an explicit trade-off. The advantages include low implementation overhead, stable execution, elimination of large model loading requirements, and compatibility with the demo and CI environments. The disadvantage is that accuracy is substantially lower than a model trained and evaluated against a manually annotated dataset. Consequently, the report does not interpret sentiment as an absolute measure of customer emotion. It is employed as a directional signal: is the proportion of negative comments abnormally elevated, how does the positive ratio evolve over time, and which brands have comment samples of sufficient size to enable trend analysis?

4. Overall Design Decisions

The initial design had a broader orientation, encompassing multiple platforms and a more comprehensive backend. During implementation, however, the team elected to narrow the scope. This reflects an important lesson from data engineering practice: broader scope does not inherently produce a superior system. If Facebook data lacks stable API access permissions, if The Coffee House's official channel ID cannot be reliably verified, and if query-based searches introduce non-official data into the warehouse, then scope expansion will diminish rather than enhance insight quality.

The final version therefore designates YouTube official channels as the clean scope. A smaller dataset with clearly defined sources, clean natural keys, and a passing quality gate is more valuable than a larger dataset whose provenance cannot be adequately explained.

Chapter III: DATA COLLECTION AND ETL PIPELINE

1. Data Sources and Extraction Strategy

The primary data source for SocialLens BI is the YouTube Data API v3. The active pages scope comprises 8 official channels.

Channel	Role in Report
Highlands Coffee Vietnam	Primary brand of focus
Trung Nguyên Legend	Primary competitor and benchmark for volume and engagement
Phúc Long	F&B competitor
Cộng Cà Phê	F&B competitor
Cheese Coffee	F&B competitor
KOI Thé Việt Nam	F&B competitor
Gong Cha Vietnam	F&B competitor
Starbucks Việt Nam	F&B competitor

The Coffee House is absent from the final warehouse because the team was unable to confirm a sufficiently reliable official YouTube channel ID. This represents a scope limitation but one that is deliberate. In BI, a single data record with an unverifiable source can corrupt an entire interpretation, particularly when share of voice or competitor benchmarks are involved.

1.1. Official Channel Extraction Strategy

The pipeline employs the following YouTube API call chain:

`channels.list -> uploads playlist -> playlistItems.list -> videos.list`

This approach is preferable to search query methods because it begins from verified official channels. Query-based searches may surface a large number of brand-related videos, but these can include reviews, reaction videos, re-uploads, news coverage,

KOL content, and material not under the brand's direct control. While such data may be valuable for broader social listening, it is unsuitable as a source of truth for an official-channel performance dashboard.

The rationale for excluding the query-search batch is discussed separately in Section 3.6, as it represents a significant data quality decision for the project.

2. Raw Data Storage

Raw data is stored in JSONL format under `data/raw/`. Raw storage serves two purposes. First, it enables auditability: when a record displays anomalous metric values, the team can trace back to the original source payload. Second, it decouples extraction from transformation: if normalization logic or sentiment classification needs to be revised, raw data can be reprocessed without necessarily re-calling the API.

Raw data is not used directly by the dashboard. It constitutes an evidential layer for auditing and reprocessing. This approach is particularly important when working with external APIs, as underlying data may change over time, comments may be deleted, and API quotas do not permit unlimited re-queries.

3. Transform — Cleansing and Normalization

Following extraction, payloads are normalized into two categories of processed data:

- `data/processed/posts.csv`
- `data/processed/comments.csv`

Timestamp fields are parsed and converted to the Asia/Ho_Chi_Minh timezone for reporting purposes. Metric fields such as views, likes, and comments are cast to numeric types and missing values are handled according to API semantics. Content type is normalized; in the final dataset, all active content records correspond to YouTube videos. Derived metrics including `engagement_count`, `engagement_rate`, and `virality_score` are computed using formulas standardized at the warehouse layer, ensuring that API responses, exports, and dashboard outputs do not diverge in their calculation logic.

4. Sentiment Analysis Pipeline

Comments are classified into positive, neutral, and negative categories, accompanied by a sentiment score. The processing workflow involves retrieving

comments from the YouTube API, applying minimal text cleaning, executing a rule-based Vietnamese sentiment fallback, and loading the results into `fact_sentiment`. The sentiment output serves a supplementary contextual role within the dashboard and is not presented as an NLP benchmark.

The pipeline also acknowledges operational constraints in YouTube comment retrieval. Certain videos may return no comments due to disabled comment sections, restricted access permissions, or 403 errors. These cases are not treated as pipeline blockers if post metadata remains valid; they are indirectly reflected through low comment coverage figures.

5. Deduplication and Upsert

A fundamental requirement of batch ETL is that repeated executions must not generate duplicate records. The project addresses this requirement through natural keys:

- Posts are deduplicated by (`platform_id`, `external_post_id`).
- Comments are deduplicated by (`platform_id`, `external_comment_id`).
- Pages are deduplicated by (`platform_id`, `external_page_id`) when a source ID is available, or by (`platform_id`, `page_name`) as a fallback.

The loader performs upsert operations into PostgreSQL rather than blind inserts. This allows repeated batches to refresh metadata or incorporate new comments without corrupting the fact table. SQL validation confirms that duplicate post natural keys and duplicate comment natural keys both equal zero.

6. Decision to Exclude the Query-Search Batch

A query-search batch was previously tested to extend coverage beyond official channels. Following review, this batch was excluded from the primary warehouse: a cleanup performed on 2026-06-01 removed 59 lower-confidence query-residual posts, 166 comments associated with those posts, and 41 orphan pages. The principal issue was that 98.76% of the reach in this batch did not originate from official pages; retaining this data would have distorted the definition of share of voice and competitor benchmarks.

The decision to exclude the query-search batch reflects the principle of prioritizing data quality over data volume. This data may be useful for social listening or

earned media analysis in future iterations, but it is incompatible with an official-channel performance dashboard. It has accordingly been archived under `data/processed_archive/official_cleanup_20260601_233646/` rather than loaded into `social_dw`.

7. Production Batch Operations

The production batch commands are standardized as follows:

```
python -m etl.cli run --sources youtube --channel-ids
$env:YOUTUBE_CHANNEL_IDS --queries= --limit 50 --comments-
limit 100 --max-search-pages 12 --database-url
$env:DATABASE_URL
```

```
python -m etl.cli quality --database-url $env:DATABASE_URL
```

```
python -m etl.cli export --database-url $env:DATABASE_URL
```

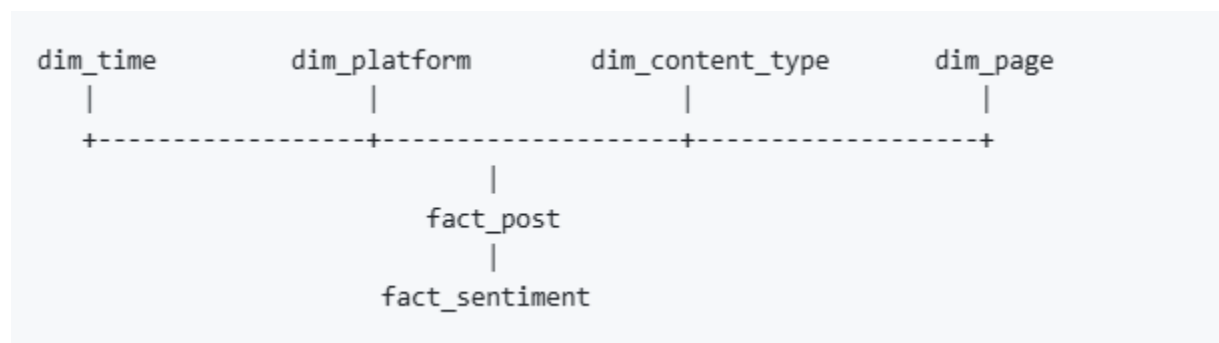
The project is not a real-time system. The dashboard reads live data from the warehouse, but the warehouse is updated only when an ETL batch is executed. This is consistent with the BI capstone objective, which calls for periodic refresh, quality validation, and snapshot-based reporting rather than streaming analytics.

Chapter IV: DATA WAREHOUSE DESIGN

1. Star Schema — Overview of the social_dw Schema

The social_dw warehouse is designed following the star schema pattern. This is a well-suited choice for BI dashboards, as the majority of analytical queries involve aggregation along dimensional axes: by date, by page, by platform, by content type, and by competitor flag. This design also establishes a clear boundary between raw/processed data and the data mart serving reporting needs.

The logical schema can be represented as follows:



`fact_post` is the central fact table for video performance. `fact_sentiment` joins to `fact_post` to enable comment-level analysis by video, page, and time period.

2. Fact Tables

`fact_post` has a grain of one video/post. Its key fields include external post ID, platform, time, content type, page, caption, URL, reach, impressions, likes, comments, shares, and saves. KPIs including `engagement_count`, `engagement_rate`, and `virality_score` are generated directly within PostgreSQL. Centralizing KPI computation in the database ensures that the API, exports, and dashboard all employ the same formula, eliminating the risk of logic divergence between the frontend and backend.

`fact_sentiment` has a grain of one comment. The table stores the external comment ID, associated post ID, platform, time, sentiment label, sentiment score, and comment text. `sentiment_ratio` is not stored directly in this fact table, as it represents an aggregated measure computed at the post, page, or date level through analytical views.

3. Dimension Tables

`dim_time` stores temporal attributes: `full_date`, `full_timestamp`, `hour_of_day`, `day_of_week`, `day_name`, `week_of_year`, `month_of_year`, `quarter_of_year`, `calendar_year`, and `is_weekend`. This layer eliminates the need for timestamp parsing at the dashboard level when constructing heatmaps or time-series charts.

`dim_platform` normalizes platform information. The final dataset contains only YouTube fact rows; however, the table retains an open design to accommodate the future addition of Facebook, TikTok, or Instagram data should API access become available.

`dim_content_type` normalizes content format classifications. In the final snapshot, all active content consists of YouTube videos. Maintaining a dedicated dimension preserves flexibility for schema extension without tightly coupling the design to YouTube specifically.

`dim_page` stores brand and channel information: external page ID, page name, industry, country code, competitor flag, and follower count where available. This is the most critical dimension for competitor benchmarking.

4. KPI Definitions in the Warehouse

The final KPI definitions are as follows:

KPI	Formula	Notes
<code>engagement_count</code>	<code>likes + comments + shares + saves</code>	Total available interactions
<code>engagement_rate</code>	<code>engagement_count / reach * 100</code>	reach is a views/impressions proxy
<code>virality_score</code>	<code>shares / reach * 100</code>	Equals 0 in this snapshot because YouTube does not provide share count

sentiment_ratio	$\frac{\text{positive_comments}}{\text{total_comments}} * 100$	Should only be interpreted in depth when the comment sample is sufficiently large
share_of_voice	$\frac{\text{page reach}}{\text{total scoped reach}} * 100$	Reach-based SOV within the official-channel scope

Notably, share_of_voice in this context is not a mention-based SOV measure across the full industry. It addresses a more constrained question: within the approved set of official YouTube channels, which brand accounts for the largest proportion of views proxy.

5. Analytical Views — 7 Views Serving the Dashboard

Seven final views serve the dashboard and export requirements:

View	Role
vw_executive_overview	Aggregate KPI snapshot: posts, reach, engagement, average rates, date range
vw_daily_engagement	Time-series reach/engagement by date and platform
vw_sentiment_trend	Sentiment by date, platform, and page
vw_content_performance	Performance by content type and competitor flag
vw_competitor_benchmark	Page/brand benchmark including SOV, engagement, and sentiment
vw_posting_time_heatmap	Performance by day of week and hour of posting
vw_viral_posts	Top videos/posts ranked by virality and engagement

These views reduce the computational load on the dashboard layer. The frontend and API do not need to re-implement complex aggregation logic; they consume data that has already been prepared at the warehouse level.

6. ETL Run Logging

The warehouse includes an `etl_runs` table to record the operational history of the pipeline. Each run captures the source, `started_at`, `finished_at`, status, number of posts and comments extracted and loaded, and an `error_message` field if a failure occurred. This table serves as an audit trail and powers the GET `/api/v1/sync/status/` endpoint, enabling the dashboard to display data freshness information rather than presenting charts disconnected from data state context.

In the final snapshot, the latest sync status is recorded from the `maintenance:official-only-cleanup` source, reflecting the decision to lock data to the official-channel scope. This constitutes important operational evidence: beyond the records in the fact tables, there is metadata that documents what the last batch operation performed.

7. Quality Gate — 43 Validation Rows

The final validation comprises 43 rows: 26 PASS, 17 INFO, and 0 FAIL. INFO rows display counts and snapshot values; PASS rows enforce mandatory conditions. Key validation checks include:

- Duplicate post natural keys: 0.
- Duplicate comment natural keys: 0.
- Orphan sentiment records: 0.
- Invalid sentiment labels: 0.
- Sentiment scores outside the range `[-1, 1]`: 0.
- Non-YouTube fact rows: 0.
- Fact rows associated with non-approved official channels: 0.
- `vw_executive_overview` reconciles with `fact_post`: PASS.

This validation layer represents what the team considers the prerequisite for "locking in" figures before writing the final report. If the quality gate were to fail, the dashboard would still render, but the report should not be submitted on the basis of those figures.

Chapter V: BACKEND API AND WEB DASHBOARD

1. Django API — Role and Design

The Django API serves as the data service layer for the dashboard. Rather than having Next.js connect directly to PostgreSQL or read from CSV files, the API exposes a stable JSON contract. This offers three advantages. First, the frontend does not require knowledge of the warehouse's physical schema. Second, warehouse-level errors can be sanitized before being returned to the client. Third, the same API can serve the web dashboard, smoke tests, and live demonstrations.

The backend uses lightweight JsonResponse rather than Django REST Framework. For a long-term production system, DRF would offer benefits in terms of serializers, pagination, schema generation, and authentication. However, within the capstone scope, the team required a small, clear, and easily testable API that primarily reads from analytical views. JsonResponse is therefore the pragmatic choice. This decision reduces complexity without altering the final data contract.

2. Endpoint Design

The primary endpoints are as follows:

Endpoint	Purpose	Primary Data Source
GET /health/	API health check	Runtime status
GET /api/v1/posts/	Paginated/limited list of posts	fact_post + dimensions
GET /api/v1/posts/{post_id}/	Detail view of a single post/video	fact_post
GET /api/v1/analytics/overview/	Aggregate KPI summary	vw_executive_overview
GET /api/v1/analytics/engagement/	Time-series engagement data	vw_daily_engagement
GET /api/v1/analytics/sentiment/	Sentiment trend data	vw_sentiment_trend

GET /api/v1/analytics/top-posts/	Top-ranked videos/posts	vw_viral_posts
GET /api/v1/analytics/content-performance/	Content performance breakdown	vw_content_performance
GET /api/v1/analytics/heatmap/	Posting time heatmap	vw_posting_time_heatmap
GET /api/v1/analytics/competitors/	Competitor benchmark	vw_competitor_benchmark
GET /api/v1/analytics/insights/	Insight summary and freshness metadata	Analytical views + sync metadata
GET /api/v1/sync/status/	Warehouse status and most recent ETL run	etl_runs + fact counts

Time-series and posts endpoints enforce bounded limit parameters to prevent oversized payloads. When the warehouse is unavailable, the API returns structured error responses in the format {detail, source_type, error_code} rather than exposing connection strings, host information, or stack traces. The smoke test conducted on 2026-06-15 confirmed that /api/v1/analytics/overview/ returns total_posts=817, total_reach=167066949, and source=warehouse.

3. Next.js Web Dashboard

The web dashboard is built using the Next.js App Router. The primary routes are:

- /dashboard: executive overview and aggregate time-series metrics.
- /content: content performance analysis.
- /sentiment: sentiment trend visualization and sentiment distribution.
- /competitors: competitive benchmark between Highlands and its competitors.
- /posts: top posts/videos table.

- /data-health: synchronization status, data freshness, and data quality indicators.

The frontend retrieves data via NEXT_PUBLIC_API_BASE_URL. Static mock JSON files have been removed from the runtime environment; the repository includes a test guard to ensure that the frontend does not import frontend/src/data/*.json. The dashboard incorporates loading and error states and supports bilingual labeling through a local i18n dictionary.

4. Dashboard Evidence and Power BI

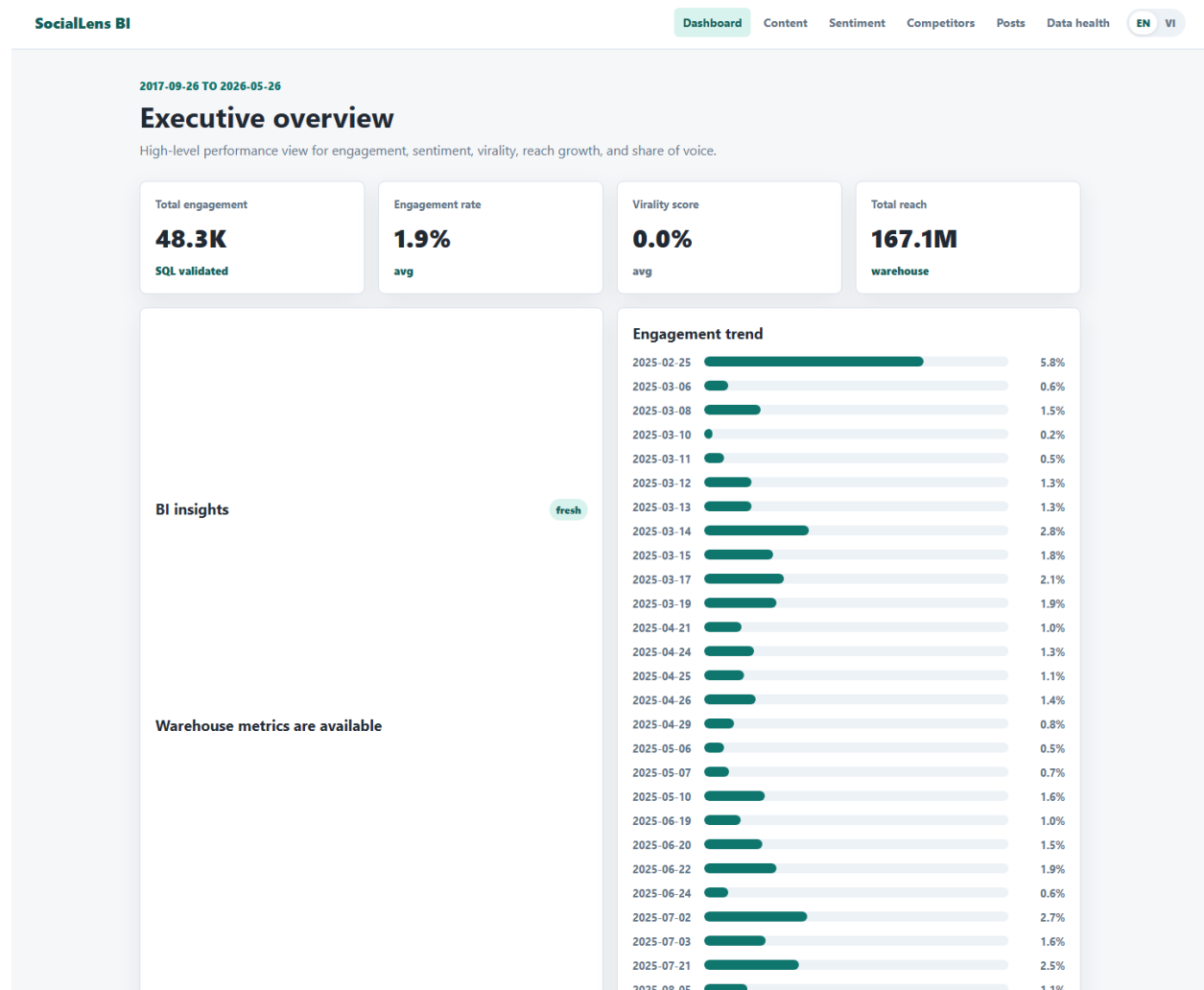


Figure 1. Executive Dashboard Overview

CONTENT PERFORMANCE

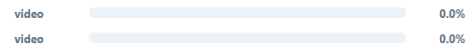
Content mix

Compare post formats by reach, engagement rate, and virality score.

Engagement rate by type



Virality by type



Content performance table

CONTENT TYPE	POSTS	REACH	ENGAGEMENT RATE	VIRALITY SCORE
video	123	160,778,238	0.7%	0.0%
video	694	6,288,711	2.1%	0.0%

Figure 2. Content Performance Dashboard

SENTIMENT ANALYSIS

Brand health

Track positive, neutral, and negative comment ratios over time.

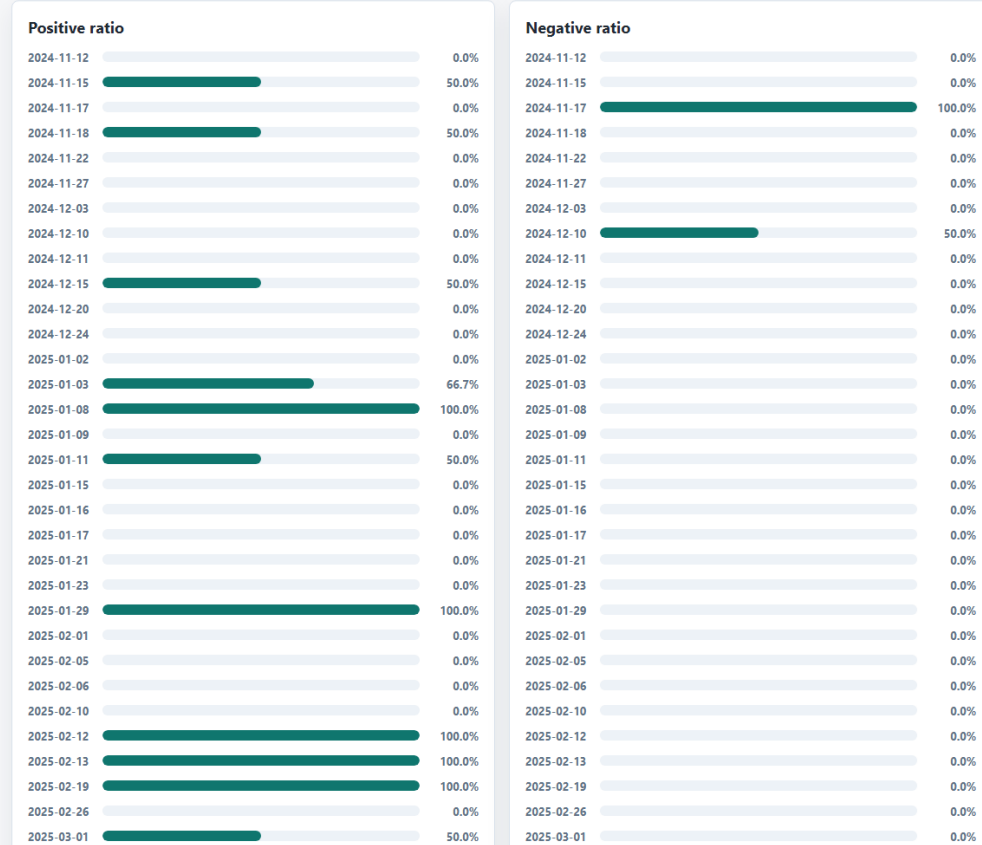
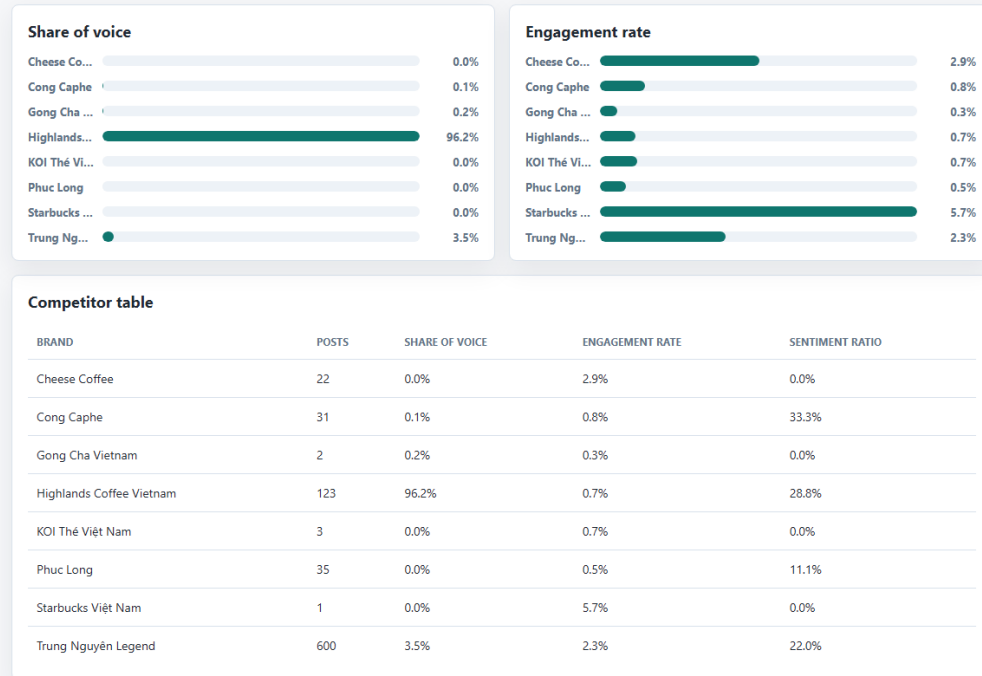


Figure 3. Sentiment Analysis Dashboard

COMPETITOR BENCHMARKING

Share of voice

Benchmark Highlands Coffee against Phuc Long and The Coffee House.

*Figure 4. Competitor Benchmark Dashboard*

POST DETAIL

Top posts

Review post-level reach, engagement count, virality score, and sentiment ratio.

Post performance table

POST	PLATFORM	BRAND	MESSAGE	VN TIME	REACH	ENGAGEMENT RATE	SENTIMENT
6d-o_3aNXNY	youtube	Phuc Long	Những khoảnh khắc chia sẻ từ khách hàng trúng giải xe ly rinh quà của Phúc Long	26 May, 13:41	8	0.0%	0.0%
D99iyDxshhQ	youtube	Phuc Long	Sinh nhật Phúc Long 58 Năm tại Flagship Phoenix	26 May, 15:17	291	0.7%	0.0%
NKnCk1Uhebw	youtube	Trung Nguyên Legend	🔥 ⚡ CÙNG G7 3IN1 KHỞI ĐẦU NGÀY MỚI TRẦN ĐẰNG LƯỢNG	25 May, 08:57	635	0.8%	0.0%
wUG91htxeE4	youtube	Trung Nguyên Legend	'21 Bài Học Cho Thế Kỷ 21': Thế Giới Trong Thời Đại Bất Định.	25 May, 13:38	225	5.8%	0.0%
klgwuxQ39U	youtube	Phuc Long	Cốc cốc... sinh nhật tới rồi cùng mở tiệc thôi 🥳 #PhucLong #HuongSacViNghê #PhucLong58Nam	19 May, 23:17	242	0.4%	0.0%
F1OE2FqQKkc	youtube	Trung Nguyên Legend	Cà phê Việt Nam "gây sốt" tại Phái bộ Gin giữ hòa bình LHQ ở Nam Sudan - QPVN	15 May, 13:37	163	4.3%	0.0%
8Vyj5HBISgE	youtube	Phuc Long	Nạp đầy 100% năng lượng tươi mới cùng "Hội Mơ Đào" 🍑 ⚡	10 May, 15:42	230	0.9%	0.0%
qDrJtn42v5Y	youtube	Phuc Long	"Biến hình" rạng rỡ cùng siêu phẩm trà Đông Phương Mỹ Nhân ✨	10 May, 15:39	373	1.1%	0.0%
MN7ljk5ewgQ	youtube	Phuc Long	MỪNG SINH NHẬT PHÚC LONG	08 May, 09:04	1,419	0.5%	0.0%
L8uaSrUJSKk	youtube	Phuc Long	Phúc Long Hùng Vương Phú Quốc xin chào	05 May, 08:37	52	1.9%	0.0%
Rs7VqPM9FdA	youtube	Phuc Long	Phúc Long Waterbus Thủ Thiêm	05 May, 08:46	181	0.6%	0.0%
ORicgleQFkM	youtube	Trung Nguyên Legend	Phương Linh, Mỹ Anh kể câu chuyện "di sản sống" cà phê tại Buôn Ma Thuột	01 May, 12:08	237	4.2%	0.0%
6x53R9tUikA	youtube	Trung Nguyên Legend	Cà phê Việt Nam trên tiến trình trở thành "Di sản sống chung của nhân loại"	27 Apr, 08:00	2,255	1.3%	0.0%
Hzk_VNIB8PM	youtube	Trung Nguyên Legend	Diễn đàn Di sản Cà phê Thế giới - Không gian đối thoại kết nối văn hoá và tri thức cà phê toàn cầu	24 Apr, 15:48	150	3.3%	0.0%

Figure 5. Top Posts / Videos Dashboard

DATA HEALTH

Pipeline status

Monitor sync source, processed row counts, and validation checks for dashboard readiness.

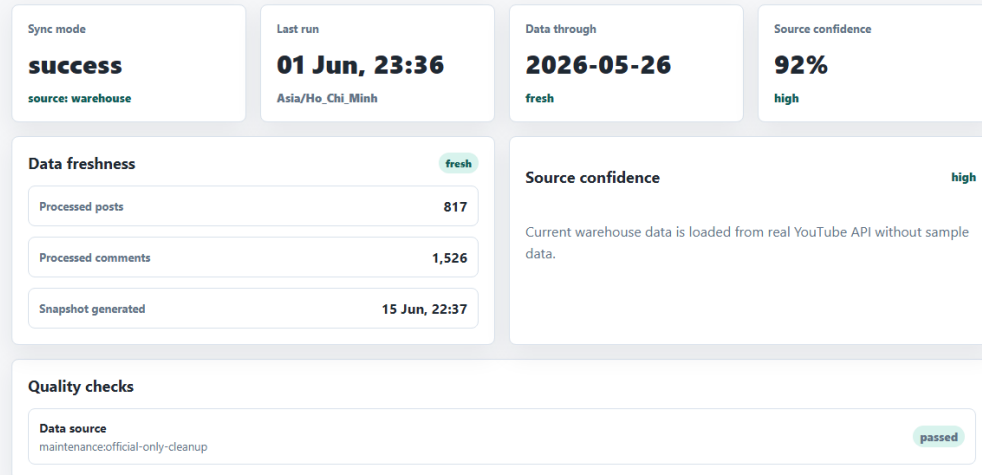


Figure 6. Data Health Dashboard

Chapter VI: ANALYTICAL RESULTS AND BI INSIGHTS

1. Data Overview

The final snapshot contains 817 videos/posts, 1,526 sentiment-classified comments, 8 official YouTube pages, and data spanning from 2017-09-26 to 2026-05-26. The total reach/views proxy is 167,066,949, total engagement is 48,325, and the average engagement rate is 1.9164%. While this is not a large-scale dataset by production BI standards, it is well-suited for a capstone BI project due to its near-nine-year historical span, sufficient brand coverage for comparative analysis, and successful passage through the quality gate.

An important caveat: all active content rows correspond to YouTube videos. Accordingly, the conclusions in this report do not apply to image posts, Facebook Stories, Reels, or TikTok short-form content. This project analyzes official YouTube channel performance and does not represent the full social media footprint of the brands under review.

2. Insight 1: Significant Imbalance Within the Dataset

The dataset is markedly imbalanced across brands. Trung Nguyên Legend accounts for 600 out of 817 posts (73.44% of all videos) and 1,258 out of 1,526 comments (82.44% of all comment sentiment records). Highlands Coffee holds 123 posts (approximately 15.06% of all videos). The remaining brands contribute substantially smaller volumes: Phúc Long with 35 posts, Cộng Cà Phê with 31, Cheese Coffee with 22, KOI Thé with 3, Gong Cha with 2, and Starbucks Việt Nam with 1.

This imbalance does not invalidate the dataset, but it governs how results must be interpreted. Absolute metrics such as post count, total engagement, and total comments will be heavily skewed toward Trung Nguyên Legend. Brands with very small sample sizes should not be subject to in-depth sentiment rankings. The report therefore prioritizes rates, proportions, confidence caveats, and context-specific benchmarks rather than direct comparisons of raw totals.

3. Insight 2: Engagement Rate Is Not Proportional to Reach Scale

Highlands Coffee Vietnam holds 123 videos but generates 160,778,238 reach/views proxy, representing 96.24% of the reach-based share of voice within the official channel set. This figure is substantial and suggests that Highlands' videos — likely

including TVC and brand awareness campaigns — achieve a level of reach far exceeding that of the remaining brands in the dataset.

However, Highlands' total engagement stands at only 10,339, considerably lower than Trung Nguyên Legend's 35,836. Highlands' average engagement rate is 0.6512%, compared to Trung Nguyên Legend's 2.2837%. This does not indicate that Highlands is underperforming; rather, it reveals that Highlands and Trung Nguyên Legend excel along two distinct KPI dimensions. Highlands leads in brand awareness, while Trung Nguyên leads in interaction cadence and comment volume.

From a marketing governance perspective, this is a significant insight. Viewing only views, Highlands appears decisively dominant. Viewing engagement, the narrative reverses. The team therefore recommends separating the dashboard and reports into two distinct KPI layers: awareness KPIs for broad-reach video content, and engagement KPIs for community-oriented content. A TVC achieving tens of millions of views proxy should not be evaluated using the same engagement rate standards applied to a community video with a substantially smaller reach.

4. Insight 3: Share of Voice Must Be Read Against Its Definition

Within the official YouTube channel scope, Highlands Coffee holds 96.24% of the reach-based share of voice. Trung Nguyên Legend follows with 3.46%, and the remaining brands collectively account for less than 1%. Viewed in isolation, the SOV chart suggests an overwhelming Highlands dominance.

However, the SOV metric here represents views proxy-based SOV, not market-wide mention-based SOV. It does not encompass Facebook, TikTok, KOL content, media coverage, user-generated content, or comments outside official channels. The correct interpretation is therefore: within the approved set of official YouTube videos, Highlands commands very substantial reach. This finding should not be interpreted as meaning Highlands accounts for 96.24% of all F&B brand discussions.

5. Insight 4: Trung Nguyên Legend Is the Strongest Benchmark for Cadence and Discussion

Trung Nguyên Legend holds 600 out of 817 videos (approximately 73.44% of all posts in the warehouse) and 1,258 out of 1,526 comments (approximately 82.44% of all comment sentiment records). It is the only brand in the dataset that

simultaneously exhibits high post volume and sufficient comment coverage to support meaningful sentiment and engagement pattern analysis.

The top engagement videos also tilt heavily toward Trung Nguyên Legend. "The Tao of Coffee - Cà Phê Đạo | Full Version VietSub" achieved 4,532 engagement interactions; "VŨ TRỤ TỈNH THỨC | TRUNG NGUYÊN LEGEND | OFFICIAL MV" achieved 4,053; and several seasonal videos such as the G7 Gold Tết campaign also recorded high engagement rates. Trung Nguyên Legend's content tends to emphasize coffee culture, brand narrative, and philosophical messaging rather than purely product-focused content.

For Highlands, the lesson is not necessarily to replicate Trung Nguyên Legend's concept, but to adopt a deeper content structure: storytelling, thematically coherent campaigns, seasonal content, and stronger calls-to-action for audience engagement. Highlands has a significant reach advantage; combining that reach with content formats that more effectively activate discussion would substantially strengthen its overall social media performance.

6. Insight 5: Virality Score of 0% Is a Data Limitation

virality_score is defined as $\text{shares} / \text{reach} * 100$; however, the YouTube Data API v3 does not provide share count in the standard API response used by this pipeline. As a result, the average virality score across the entire snapshot is 0.0000%. This is not evidence that videos are not being shared; it simply indicates that share data is unavailable from the current data source.

In the report, virality score should be retained as a defined KPI but accompanied by a clear caveat. Maintaining this metric has design value because the schema and dashboard are already structured to accommodate it once a reliable share or referral analytics source becomes available — for example, the YouTube Analytics API with channel ownership permissions, or a third-party social listening tool. In the current version, virality score should not be used for content ranking purposes.

7. Insight 6: Sentiment Is Safe but Not Strongly Positive

A total of 1,526 comments were sentiment-classified: 351 positive, 1,148 neutral, and 27 negative. The positive ratio is 23.00% and the negative ratio is 1.77%. From a risk monitoring perspective, this represents a relatively safe signal. Negative

sentiment is not elevated and does not indicate a sentiment crisis within the snapshot period.

From a brand health perspective, however, the disproportionately large neutral ratio suggests that the majority of comments do not express strong positive sentiment. Viewers may primarily be commenting with informational responses, tagging friends, posing questions, or leaving brief acknowledgments. For an F&B brand where emotional experience and consumption habits play a significant role, the goal is not only to minimize negative sentiment but also to actively increase positive comment volume.

For an F&B brand, this neutral-heavy pattern is important. Coffee and beverage content is not only expected to inform; it is expected to trigger appetite, routine, memory, social moments, or store experience. If most comments remain neutral, the content may be reaching viewers but not consistently activating emotional connection. For Highlands, the practical implication is not only to reduce negative comments, but to design more content that invites personal reactions: product rituals, seasonal moments, store experience, customer stories, and clearer calls-to-action. In other words, sentiment should be read as a signal of emotional activation, not merely as a risk indicator.

Highlands records a sentiment ratio of 28.75%, higher than Trung Nguyên Legend's 22.02%; however, Highlands has only 240 comments. Cộng Cà Phê shows a sentiment ratio of 33.33% but with only 12 comments. KOI Thé has 0 comments, Gong Cha has 4, and Starbucks has 1. The report therefore refrains from ranking sentiment in depth for brands with small samples. The correct reading is: overall sentiment is safe; Highlands and Trung Nguyên Legend have sufficient data for monitoring; the remaining brands require additional data before conclusions can be drawn.

8. Insight 7: High-Reach Videos and High-Engagement Videos Are Distinct Groups

The top engagement videos in the export dataset include:

Video	Page	Engagement	Engagement Rate
--------------	-------------	-------------------	------------------------

The Tao of Coffee - Cà Phê Đạo	Trung Nguyên Legend	4,532	1.8160%
VŨ TRỤ TỈNH THỨC	Trung Nguyên Legend	4,053	0.7345%
HIGHLANDS COFFEE X HUỖNH LẬP	Highlands Coffee Vietnam	4,006	1.4513%
Cà phê G7 Gold	Trung Nguyên Legend	2,183	3.0674%
Bloomberg: Trung Nguyên Legend's Vision	Trung Nguyên Legend	1,541	0.8875%

In contrast, several high-reach Highlands videos achieve tens of millions of views proxy while recording very low engagement rates. This is a commonly observed phenomenon with awareness-oriented or paid/boosted content: a very large denominator in reach suppresses the engagement rate, even when the content achieves significant media value if the primary objective is brand exposure.

This leads to a technical dashboard recommendation: the top posts view should not rely on a single ranking criterion. At minimum, two perspectives should be provided — top by reach and top by engagement or engagement rate. Ranking solely by engagement rate may over-rank low-reach videos; ranking solely by reach may obscure videos that drive meaningful community interaction.

9. Insight 8: The Posting Time Heatmap Should Be Read Through Both Volume and Efficiency Lenses

vw_posting_time_heatmap contains 122 day/hour cells. When sorted by total engagement, several time slots stand out: Monday 10:00 with 5,010 engagement across 19 posts, Wednesday 20:00 with 4,297 engagement across 14 posts, and Thursday 19:00 with 4,272 engagement across 7 posts. These slots have sufficient volume and high aggregate engagement to warrant inclusion in any experimentation shortlist.

When sorted by average engagement rate alone, several slots record very high values — for example, Friday 23:00 at 6.5477% but from only 2 posts, and Sunday 21:00 at 5.7554% from only 1 post. These slots are interesting but lack sufficient data to be codified into a posting schedule. They are appropriate for generating hypotheses, not for drawing conclusions.

Accordingly, posting schedules should be tested over a 4-to-6-week period on a per-brand basis, rather than treating the current heatmap as a fixed scheduling template. For Highlands, the team recommends prioritizing time slots that exhibit both meaningful volume and notable aggregate engagement — specifically Monday 10:00, Wednesday 20:00, and Thursday 19:00/20:00 — followed by measurement of engagement rate and comment quality outcomes.

10. Insight 9: Data Quality Is a Project Output, Not Merely a Preparatory Step

One dimension easily overlooked in BI reporting is data quality. In SocialLens BI, data quality itself constitutes a reportable outcome. The query-search batch was excluded because 98.76% of its reach did not originate from official pages. Following the cleanup, the warehouse retains 817 official posts and 1,526 official comments. The quality validation returns 0 FAIL. The frontend no longer falls back to static JSON. The API and dashboard read exclusively from PostgreSQL.

This has practical significance. If the dashboard asserts that "Highlands accounts for 96.24% share of voice," a reviewer may reasonably ask: share of voice within which scope? Does the source data include non-official videos? Are there duplicates? Are there orphan comment records? This report is able to answer those questions with reference to the schema design, validation SQL, and cleanup evidence.

11. Dashboard and Insight Visualization

The dashboard is designed to serve readers who need rapid orientation while maintaining full analytical traceability. The executive page displays aggregate KPIs: total posts, total reach, total engagement, average engagement rate, date range, and data source. This page is intended for management stakeholders requiring a situational overview within a few minutes.

The content performance page focuses on the question of which content formats and messaging approaches perform effectively. In this snapshot, all content consists of

video, so the page primarily compares the primary brand against the competitor group. Upon expansion to a multi-platform scope, this view could be used to compare video, reels, images, stories, and livestream formats.

The sentiment page displays positive, neutral, and negative distributions over time and by page. Given that sentiment is produced by a rule-based fallback, the dashboard should present it as a directional signal. Sentiment charts should not be used as the sole evidence for conclusions regarding brand affinity or customer satisfaction.

The competitors page is the most strategically significant page. It illustrates Highlands' commanding reach advantage alongside Trung Nguyên Legend's superior content volume and engagement. Appropriate chart types include bar charts for share of voice, engagement, and post count, complemented by a detailed page-level benchmark table.

The posts page enables review of individual top-performing videos ranked by engagement or virality. This page bridges the gap between aggregate KPIs and specific content artifacts. When preparing reports or marketing proposals, the team can draw on top engagement videos to analyze content format, messaging, and contextual factors.

The data health page presents data source information, freshness status, counts, and synchronization status. For a BI project, this page is particularly important as it communicates whether the dashboard is reading live data from the warehouse. In the smoke test conducted on 2026-06-15, the API overview returned `source=warehouse` and `total_posts=817`; the sync status indicated the latest source as `maintenance:official-only-cleanup`.

Chapter VII: TESTING AND QUALITY ASSURANCE

1. Multi-Layer Testing Strategy and Warehouse Validation

Warehouse validation constitutes the most critical layer. SQL validation in `warehouse/queries/dashboard_validation.sql` checks counts, natural keys, fact-dimension referential integrity, domain values, and KPI reconciliation. The final result: 43 validation rows, 26 PASS, 17 INFO, and 0 FAIL.

These checks surface errors that a dashboard would have difficulty detecting. For example, a dashboard may still render in the presence of duplicate posts, but total engagement figures would be inflated. A dashboard may still draw sentiment charts in the presence of orphan comments, but per-page benchmarks would be incorrect. Warehouse validation is therefore executed prior to export generation and screenshot capture.

2. Python Regression Tests

Python tests cover the ETL pipeline, extractors, normalization logic, quality checks, API endpoints, warehouse SQL, and final delivery artifacts. After introducing a coverage gate, `python -m pytest -q` yields 59 passed, 1 skipped, and a total coverage of 70.59%, exceeding the 60% threshold. The optional live YouTube test executes only when an API key and quota are available.

Test coverage should not be interpreted as an absolute guarantee of correctness, but it facilitates detection of regressions in core components. In particular, API tests confirm that the response shape adheres to the SPEC.md lightweight specification, and final delivery tests confirm that exports exist and that the Data Dictionary documents all final views.

3. Frontend Tests, Linting, and Build

The frontend employs Jest/React Testing Library tests, ESLint flat config linting, and Next.js production builds. Current results:

- `npm run lint`: pass.
- `npm test -- --runInBand`: 7 tests pass; coverage baseline pass.
- `npm run build`: pass.
- `npm audit --omit=dev --audit-level=moderate`: 0 vulnerabilities.

Additionally, the repository includes a static guard to ensure the frontend runtime does not import mock JSON. This is a small but consequential test: a dashboard that appears correct while reading fallback data represents a serious deficiency in a BI context.

4. API Smoke Test

The API smoke test validates the `/health/`, `/api/v1/analytics/overview/`, and `/api/v1/sync/status/` endpoints. When the project was started on 2026-06-15, the API returned:

- Health: 200 OK.
- Overview: `total_posts=817`, `total_reach=167066949`, `source=warehouse`.
- Sync status: `warehouse counts posts=817`, `comments=1526`.

All dashboard routes — `/dashboard`, `/content`, `/sentiment`, `/competitors`, `/posts`, and `/data-health` — returned HTTP 200.

Chapter VIII: LIMITATIONS AND FUTURE DEVELOPMENT

1. YouTube-Only Scope

The most significant limitation is the YouTube-only scope. Conclusions drawn in this report are representative of official YouTube channels only and do not reflect the full social media performance of Highlands or the broader F&B sector. Facebook, Instagram, TikTok, earned media, KOL content, and website campaign data all fall outside the current snapshot.

The appropriate path for future development is not simply to add all available sources. The correct approach is controlled expansion: each new data source must have a defined source confidence rating, an official/non-official flag, clearly specified natural keys, and a dedicated quality gate. Without these controls, the warehouse would replicate the issues encountered with the query-search batch.

2. Reach as a Proxy Metric

The YouTube API in this pipeline provides a views/impressions proxy rather than unique reach as defined by marketing platforms. Accordingly, the report uses the term reach/views proxy. When presenting findings to non-technical audiences, it is important to clarify that "reach" in the dashboard refers to view counts or impressions proxy, not unique individual users.

In future iterations, access to the YouTube Analytics API or brand-owned internal data could enable the incorporation of watch time, average view duration, subscriber gain, impressions click-through rate, and unique viewer counts. These metrics would substantially deepen the YouTube dashboard beyond what views, likes, and comments alone can reveal.

3. Virality Score at 0%

$\text{virality_score} = \text{shares} / \text{reach} * 100$; however, the YouTube API does not expose share count in the pipeline's current implementation. Consequently, the virality score across the entire final snapshot is 0.0000%. This is a data limitation, not a conclusion that videos are not being shared. The report should avoid providing substantive interpretation of this metric until a reliable share, referral, or share URL analytics source is available.

4. Rule-Based Sentiment Classification

Sentiment classification currently relies on a lightweight Vietnamese rule-based fallback. The advantage is that it is computationally lean, stable, and executable within the capstone environment. The disadvantage is that it is not sufficiently accurate to replace a Vietnamese NLP model evaluated against a standardized benchmark. A concrete path for future development would involve constructing a manually annotated comment dataset, fine-tuning or benchmarking PhoBERT or ViSoBERT, and subsequently comparing precision and recall by sentiment label.

Beyond sentiment polarity, the project could incorporate topic extraction: covering product-related, pricing, ambiance, service, promotional, delivery, and in-store experience categories. This represents a substantially higher-value analytical layer for F&B brands, as negative sentiment is only actionable when paired with knowledge of what the negativity pertains to.

5. Absence of The Coffee House

The Coffee House is not included in the clean warehouse because a sufficiently reliable official YouTube channel ID could not be verified. This represents a gap in the competitor set; however, the decision to exclude it is methodologically sound. In BI, including a competitor with an unverifiable data source has the potential to produce greater benchmark distortion than omitting it entirely.

The recommended path forward is to verify the official channel through the brand's own website, social media links, or brand documentation. The channel ID should be added to YOUTUBE_CHANNEL_IDS and loaded into the warehouse only once it has been confirmed to a sufficient level of confidence.

Chapter IX: CONCLUSION

1. Revisiting the Original Business Questions

SocialLens BI has addressed the majority of its original business questions within the defined scope. The project successfully constructed an end-to-end pipeline from real YouTube official channel data to a PostgreSQL warehouse, complete with quality gates, a REST API, a web dashboard, and export artifacts for reporting purposes. More significantly, the project did not merely produce a dashboard; it established a defensible data logic: why this data source was selected, why others were excluded, how KPIs are computed, whether duplicates exist, and whether views reconcile with fact table records.

From a business standpoint, the report surfaces a noteworthy paradox: Highlands Coffee overwhelmingly leads in reach/views, while Trung Nguyên Legend outperforms in content volume, engagement, and comment coverage. This prevents a simplistic conclusion that the brand with the highest view count is the brand executing the most effective social media strategy. Within YouTube official-channel BI, brand awareness and audience engagement must be measured using two distinct KPI frameworks.

2. Academic and Technical Value of the Process

The project also demonstrates the foundational role of data engineering within BI practice. Without official-channel extraction, deduplication, star schema design, generated KPIs, validation SQL, and cleanup batch operations, the dashboard could easily become a visually polished layer over noisy data. The team chose the opposite approach: a smaller but cleaner dataset that is more interpretable and easier to defend.

From an academic perspective, the project validates the principle that BI concepts taught in the classroom are inseparable from practical implementation. The grain of a fact table determines how aggregation behaves; source confidence determines the reliability of benchmarks; quality gates determine whether figures can be finalized; and a dashboard is meaningful only to the extent that its readers understand the KPI definitions underlying it.

3. Recommended Next Steps If the Project Continues

Several questions remain outside the current scope. The project does not represent the full social media landscape due to the absence of Facebook, TikTok, and Instagram data. Sentiment classification is not a high-performance NLP model. Virality score is not meaningful due to the absence of share count data. The Coffee House has not been incorporated into the benchmark. However, these limitations are explicitly documented, have technical justifications, and have concrete improvement pathways identified. The main lesson from SocialLens BI is clear: the brand with the largest reach is not automatically the brand with the strongest engagement. Highlands Coffee dominates awareness in this YouTube dataset, while Trung Nguyễn Legend sets the stronger benchmark for cadence, discussion, and interaction. That contrast is exactly why a BI system is useful: it prevents a single large metric from becoming the whole story.

APPENDICES

A. Abbreviated Data Dictionary

Table/View	Grain	Role
dim_time	A single reporting timestamp	Analysis by day, hour, week, and month
dim_platform	A single platform	Platform normalization; final data is YouTube only
dim_content_type	A single content format	Normalization of video/reel/livestream/etc.
dim_page	A single official page/channel	Brand and competitor lookup
fact_post	A single video/post	Video performance: reach, engagement, KPIs
fact_sentiment	A single comment	Sentiment label and score per comment
vw_executive_overview	A single snapshot	Aggregate KPI summary
vw_daily_engagement	Date + platform	Time-series engagement
vw_sentiment_trend	Date + platform + page	Sentiment trend
vw_content_performance	Platform + content type + competitor flag	Content performance breakdown
vw_competitor_benchmark	Platform + page	Brand benchmark
vw_posting_time_heatmap	Day + hour + platform	Posting time heatmap
vw_viral_posts	A single video/post	Top video/post table

B. API Endpoint List

GET /health/

GET /api/v1/posts/

GET /api/v1/posts/{post_id}/

GET /api/v1/analytics/overview/

GET /api/v1/analytics/engagement/

GET /api/v1/analytics/sentiment/

GET /api/v1/analytics/top-posts/

GET /api/v1/analytics/content-performance/

GET /api/v1/analytics/heatmap/

GET /api/v1/analytics/competitors/

GET /api/v1/analytics/insights/

GET /api/v1/sync/status/

C. Consolidated Test Results

Category	Command	Final Result
Python tests + coverage	python -m pytest -q	59 passed, 1 skipped, coverage 70.59%
Django configuration	python backend\manage.py check	Pass, no issues
Warehouse quality gate	python -m etl.cli quality --database-url \$env:DATABASE_URL	43 rows, 26 PASS, 17 INFO, 0 FAIL
Frontend tests	npm test -- --runInBand	7 passed

Frontend production build	npm run build	Pass
Frontend lint	npm run lint	Pass
Production dependency audit	npm audit --omit=dev --audit-level=moderate	0 vulnerabilities
API smoke test	/health/, /analytics/overview/, /sync/status/	200 OK, source=warehouse

D. Validation Summary and Final Data Snapshot

Category	Result
Warehouse validation rows	43
PASS	26
INFO	17
FAIL	0
Fact posts	817
Fact sentiment comments	1,526
Active pages	8
Date range	2017-09-26 to 2026-05-26
Total reach/views proxy	167,066,949
Total engagement	48,325
Average engagement rate	1.9164%
Average virality score	0.0000%